If we successfully address this question, we will have made significant headway toward showing the world we would expect to see if Many-Worlds were true is the world we actually do see. That is, a world that is closely approximated by classical physics, except for quantum measurement events, during which the probability of obtaining any particular outcome is given by the Born rule.

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The issue of probabilities is often phrased as trying to derive why probabilities are given by amplitudes squared. But that's not really the hard part. Squaring amplitudes in order to get probabilities is a very natural thing to do; there weren't any worries that it might have been the wave function to the fifth power or anything like that. We learned that back in Chapter Five, when we used qubits to explain that the wave function can be thought of as a vector. That vector is like the hypotenuse of a right triangle, and the individual amplitudes are like the shorter sides of that triangle. The length of the vectors equals one, and by Pythagoras's theorem that's the sum of the squares of all the amplitudes. So "amplitudes squared" naturally look like probabilities: they're positive numbers that add up to one.

The deeper issue is why there is anything unpredictable about Everettian quantum mechanics at all, and if so, why there is any specific rule for attaching probabilities. In Many-Worlds, if you know the wave function at one moment in time, you can figure out precisely what it's going to be at any other time, just by solving the Schrödinger equation. There's nothing chancy about it. So how in the world is such a picture supposed to recover the reality of our observations, where the decay of a nucleus or the measurement of a spin seems irreducibly random?

Consider our favorite example of measuring the spin of an electron. Let's say we start the electron in an equal superposition of spin-up and spin-down with respect to the vertical axis, and send it through a Stern-Gerlach magnet. Textbook quantum mechanics says that we have a 50 percent chance of the wave function collapsing to spin-up, and a 50 percent chance of it collapsing to spin-down. Many-Worlds, on the other hand, says there is a 100 percent chance of the wave function of the universe evolving from one world into two. True, in one of those worlds the experimenter will have